Formalização do Modelo v2

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2026-04-19

1 Environment

Definition 1 (Institutional Design Game). *The game $\Gamma(N, p, r, \alpha, \beta, c)$ has the following primitives.*

- *There are $N \geq 3$ players: one hegemon H and $n = N - 1$ weak states $W_1, \dots, W_n$.*
- *Nature draws $\theta \in \{0, 1\}$ with prior $\Pr(\theta = 1) = p \in (0, 1)$.*
- *The state determines the value of multilateral cooperation: $V(0) = 1$ and $V(1) = r > 1$.*
- *The hegemon observes θ; the weak states do not.*
- *If bargaining fails (disagreement):*
  - *Each weak state receives 0.*
  - *The hegemon receives $\alpha V(\theta)$, where $\alpha \in (0, 1/r)$, representing bilateral alternatives proportional to the value of cooperation.*
- *The common discount factor is $\beta \in (0, 1)$.*
- *Each weak state pays an entry cost $c > 0$ to join the institution.*

Let

$$V_e(\mu) \equiv 1 + \mu(r - 1)$$

denote the expected value of cooperation at posterior belief μ.

Remark 1 (Normalization). *The assumption $d_W = 0$ is without loss of generality. It requires only that (i) weak states' outside options are strictly lower than the hegemon's, and (ii) outside options are symmetric across weak states. Given these conditions, we redefine the pie as the surplus above the weak states' common outside option, which normalizes their disagreement payoff to zero.*

Remark 2 (Bilateral alternatives)**.** *The hegemon's disagreement payoff* $\alpha V(\theta)$ *captures bilateral alternatives that are proportional to the value of multilateral cooperation. The same factors that make multilateral cooperation valuable (favorable trade environment, complementary economies) also improve bilateral alternatives. The parameter* $\alpha < 1/r$ *ensures that bilateral alternatives are strictly dominated by multilateral cooperation for all* θ*.*

2 Institutional Packages

Definition 2 (Institutional Packages)**.** *The analysis compares two institutional packages that differ only in the voting rule. Both packages have symmetric proposal rights (random proposer with probability* $1/N$ *each).*

- ***Package M (Majority)****: decisions pass with strict majority support (*$q = \lfloor N/2 \rfloor + 1$ *votes).*
- ***Package U (Unanimity/Consensus)****: decisions require unanimous support (*$q = N$ *votes).*

Remark 3 (Why not agenda control?)**.** *By agenda control we mean here monopoly over proposal rights (exclusive proposer). Under either voting rule, if the hegemon were the exclusive proposer and* $d_W = 0$*, weak states' expected bargaining payoff would be zero. No weak state would pay the entry cost* $c > 0$*, and the institution would never form. Monopoly proposal power is therefore self-defeating. The relevant institutional choice is between voting rules, holding proposal rights symmetric. Other forms of agenda influence—such as shaping the set of feasible proposals—are discussed in the extension.*

3 Timing

The game has three stages.

1. **Stage 0 (Institutional choice).** The hegemon chooses the voting rule $R \in \{M, U\}$.
2. **Stage 1 (Information and entry).** Nature draws θ. The hegemon observes θ and commits to a public signal structure $\pi : \{0, 1\} \to \Delta(S)$. Weak states observe the realized signal $s \in S$, update to posterior $\mu = \Pr(\theta = 1 \mid s)$, and simultaneously decide whether to enter (paying cost c).
3. **Stage 2 (Bargaining).** If the institution forms, members play a two-round Baron-Ferejohn bargaining game over a pie of size $V(\theta)$:
    - **Round 1**: a proposer is selected uniformly at random (probability $1/N$). The proposer

offers a division. All members vote according to rule R. If accepted, the pie is divided as proposed. If rejected, the game proceeds to Round 2.

- **Round 2 (terminal)**: a new proposer is selected uniformly at random. The proposer offers a division. All members vote. If accepted, the pie is divided. If rejected, each player receives their disagreement payoff (0 for each W, $\alpha V(\theta)$ for H).

4 Solution Concept

The solution concept is Perfect Bayesian Equilibrium (PBE). Strategies are sequentially rational given beliefs, and beliefs are updated by Bayes' rule on the equilibrium path. Off-path beliefs satisfy sequential rationality.

5 Game Tree

The game tree is presented for the unanimity subgame (Package U). Under Package M (majority), the structure is identical except that W proposers exclude H from the coalition, eliminating the screening problem.

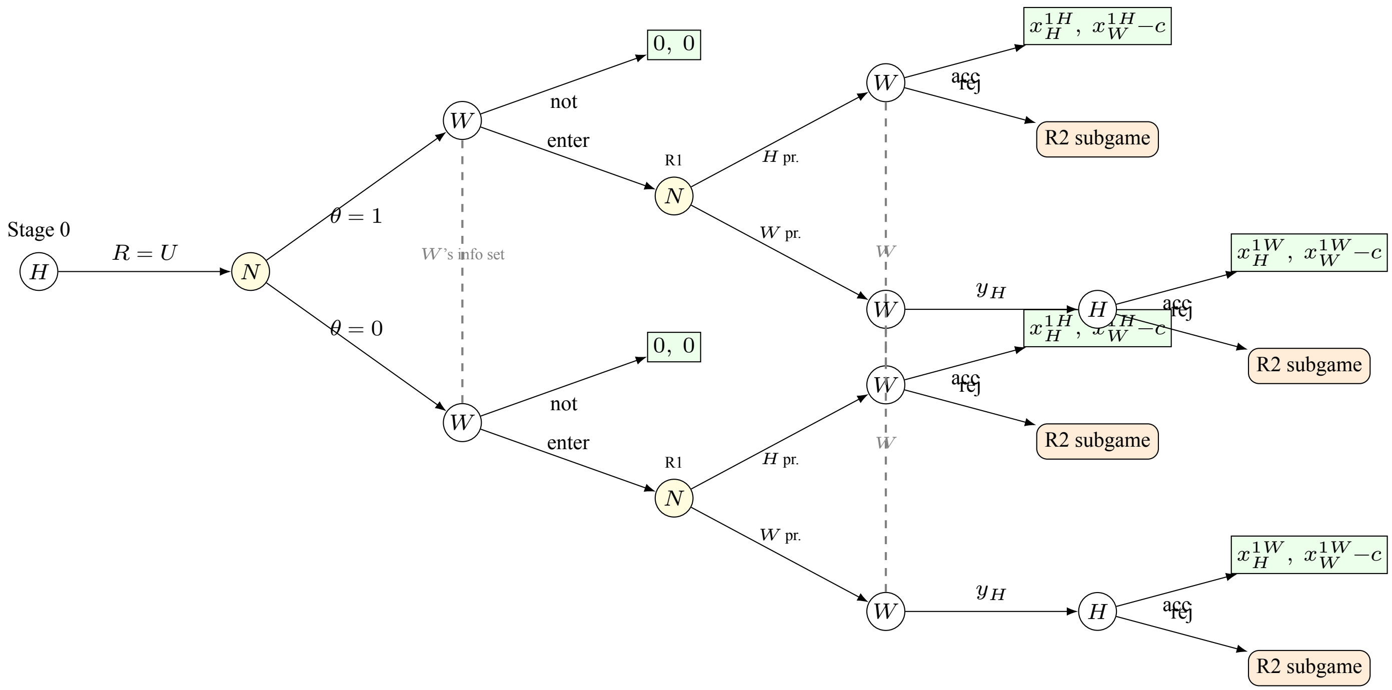

Figure 1: Main game tree under Package U (unanimity): Stage 0 through Round 1. H chooses R; Nature draws θ; H sends signal (not shown); W forms posterior μ and decides entry (paying c only if entering). In R1, Nature selects proposer ($1/N$). When H proposes, W votes to accept or reject. When W proposes, W offers y_H to H, who accepts or rejects. Any rejection in R1 leads to the R2 subgame (Figure 2). Dashed lines: W's information sets (W does not observe θ). Under Package M, W proposers exclude H, eliminating the screening branch.

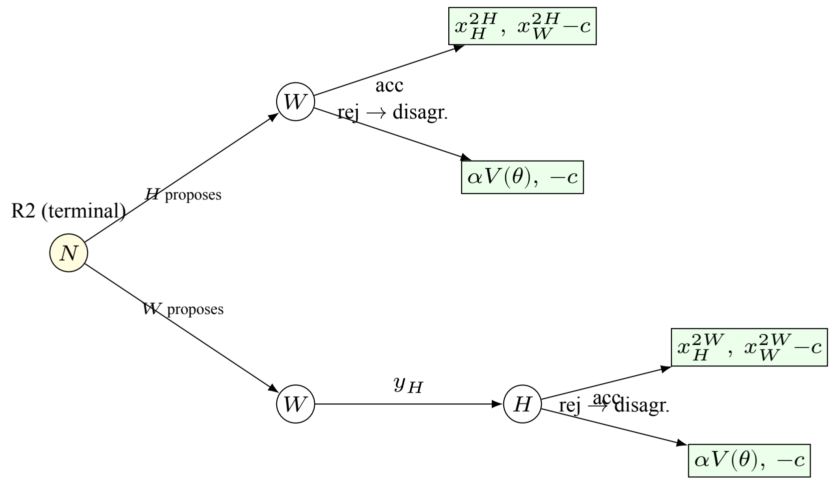

Figure 2: R2 subgame (terminal round) under unanimity. Nature selects proposer. When W proposes, W offers y_H to H; H accepts or rejects. When H proposes, W accepts or rejects. Rejection leads to disagreement payoffs: $\alpha V(\theta)$ for H, $-c$ for W (entry cost sunk). This subgame is reached after any rejection in R1 (Figure 1). W's information set spans both θ branches (not shown here for clarity; see Figure 1).

6 Equilibrium Analysis (to be developed)

6.1 Terminal Round (R2) under Unanimity

When H proposes (prob $1/N$): offers 0 to each W ($d_W = 0$), all accept. H keeps $V(\theta)$.

When W_j proposes (prob $1/N$): offers 0 to other W's. Faces screening problem for H:

- **Aggressive offer**: $y_H = \alpha$ ($= \alpha V(0)$). Type $\theta = 0$ accepts ($\alpha \geq \alpha$). Type $\theta = 1$ rejects ($\alpha < \alpha r$), receives $\alpha V(1) = \alpha r$.
- **Conservative offer**: $y_H = \alpha r$ ($= \alpha V(1)$). Both types accept.

Expected payoff for W as proposer:

$$F^{agg}(\mu) = (1-\mu)(1-\alpha) \tag{1}$$
$$F^{con}(\mu) = V_e(\mu) - \alpha r \tag{2}$$

Screening cutoff: W prefers aggressive when $F^{agg}(\mu) > F^{con}(\mu)$:

$$\mu_s = \frac{\alpha(r-1)}{r-\alpha} \tag{3}$$

Closed form. W plays aggressive if $\mu < \mu_s$, conservative if $\mu > \mu_s$.

Continuation values R2:

$V_W^{R2}(\mu)$:

$$\mu < \mu_s : \quad \frac{(1-\mu)(1-\alpha)}{N} \tag{4}$$
$$\mu > \mu_s : \quad \frac{V_e(\mu) - \alpha r}{N} \tag{5}$$

Continuity at μ_s: to verify.

$V_H^{R2}(\theta, \mu)$:

$$\theta = 1 : \quad \frac{r[1 + (N-1)\alpha]}{N} \quad \text{(constant in } \mu\text{)} \tag{6}$$
$$\theta = 0,\ \mu < \mu_s : \quad \frac{1 + (N-1)\alpha}{N} \tag{7}$$
$$\theta = 0,\ \mu > \mu_s : \quad \frac{1 + (N-1)\alpha r}{N} \quad \text{(overpaid)} \tag{8}$$

Jump in $E[V_H^{R2}(\mu)]$ at cutoff μ_s:

$$\text{Jump} = \frac{(N-1)\alpha(r-1)(1-\mu_s)}{N} > 0 \tag{9}$$

6.2 Terminal Round (R2) under Majority

When H proposes (prob $1/N$): buys $q-1$ votes (offers 0, $d_W = 0$). Keeps $V(\theta)$.

When W proposes (prob $(N-1)/N$): excludes H (more expensive), forms coalition with $q-1$ other W's (offers 0). Keeps $V(\theta)$. H receives $\alpha V(\theta)$ (bilateral alternative, excluded).

$$V_H^{R2}(\theta, M) = \frac{V(\theta)[1+(N-1)\alpha]}{N} \quad \text{— linear in } \mu\text{, no screening.} \tag{10}$$

$$V_W^{R2}(\mu, M) = \frac{V_e(\mu)}{N} \quad \text{[to verify]} \tag{11}$$

6.3 Round 1 (R1) — to derive

Under unanimity: screening works in R1 because $V(\theta)$ creates a structural gap $(r-1)/N$ between types when H proposes. Type $\theta = 0$ cannot replicate $\theta = 1$'s payoff by bluffing.

IC constraints in R1:

$$\theta = 0 \text{ accepts}: \quad y_H \geq \frac{\beta[1+(N-1)\alpha r]}{N} \tag{12}$$

$$\theta = 1 \text{ rejects}: \quad y_H < \frac{\beta \cdot r[1+(N-1)\alpha]}{N} \tag{13}$$

$$\text{Gap}: \quad \frac{\beta(r-1)}{N} > 0 \quad \checkmark \tag{14}$$

Under majority: no screening (W excludes H). Linear.

6.4 Entry — to derive

6.5 Bayesian Persuasion — to derive

6.6 Institutional Comparison — to derive

7 Open Questions

1. **Direction of entry without direct benefit**: under unanimity (aggressive branch), V_W is decreasing in μ — entry easier at low μ. Under majority, V_W is increasing — entry easier at high μ. Opposite directions. Derive consequences for BP.
2. **Feasibility of absolute offers**: W proposes amounts without knowing $V(\theta)$. Conservative offer $y_H = \alpha r$: feasible when $\theta = 0$ ($\alpha r < 1$ since $\alpha < 1/r$)? Yes. ✓
3. V_W **under majority**: when W proposes and excludes H, W keeps $V(\theta)$ but does not know $V(\theta)$. W proposes amounts and receives residual $V(\theta) -$ offers. Residual depends on θ.
4. **Continuity of V_W^{R2} at μ_s**: verify algebraically.
5. **Condition for H to prefer unanimity**: when does screening channel + superior extraction outweigh entry disadvantage?